DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

Instructor: Prof. Neelesh Upadhye

Lecture 7: Mean-Variance Analysis

Portfolio Return & Risk · The Covariance Matrix · Diversification

August 6, 2026

Today's Agenda

1. Why combine assets at all?

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2. The two-asset portfolio: return and variance
3. The role of correlation
4. Many assets: the matrix formulation
5. Diversification as N grows: systematic vs. unsystematic risk

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- Read and interpret a covariance matrix.
- Explain why diversification reduces risk, and distinguish systematic from unsystematic risk.

The Umbrella Seller and the Ice-Cream Seller

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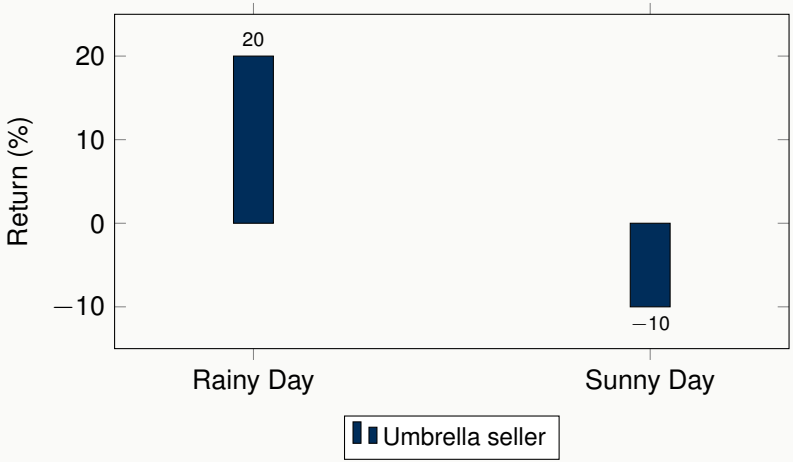
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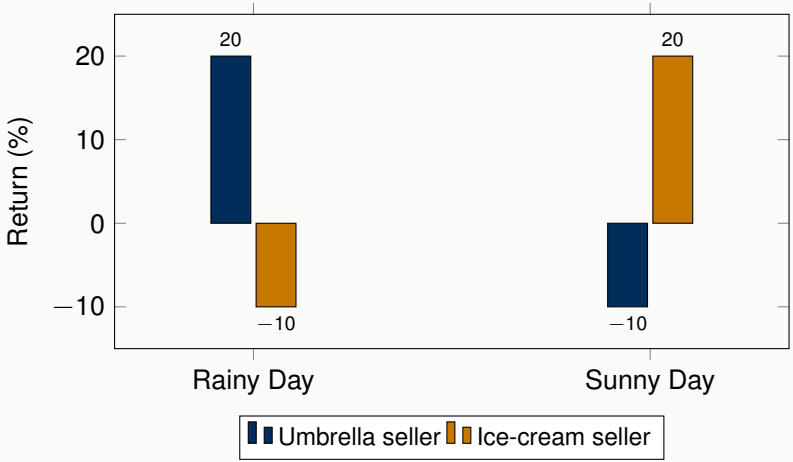
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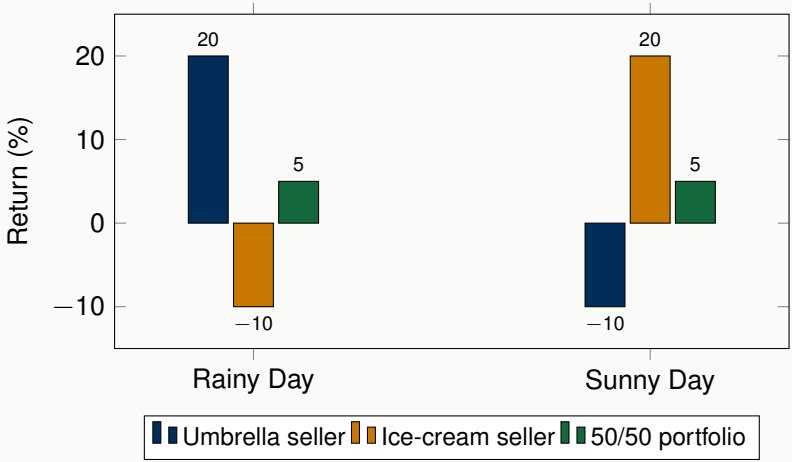
Diversification, Drawn



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Portfolio Return: Always a Weighted Average

Definition

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But risk is a different story

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \rho_{12} \sigma_1 \sigma_2$$

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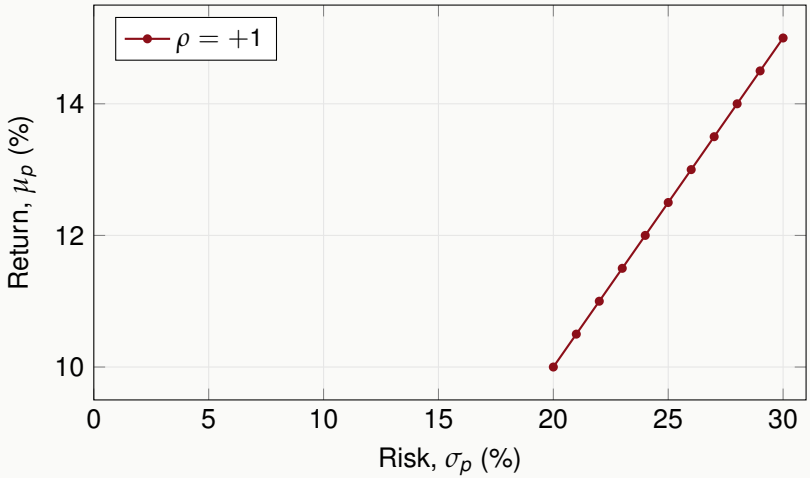
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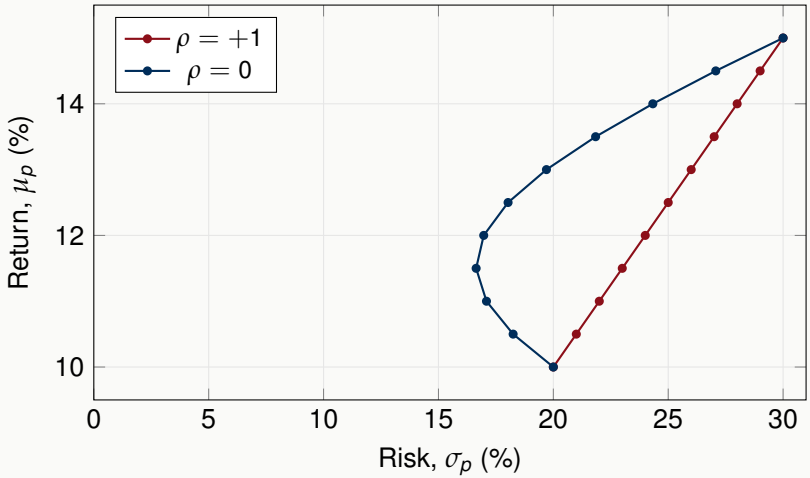
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$\sigma_p \approx 19.35\%$ – **less** than the weighted average 24%: diversification already at work.

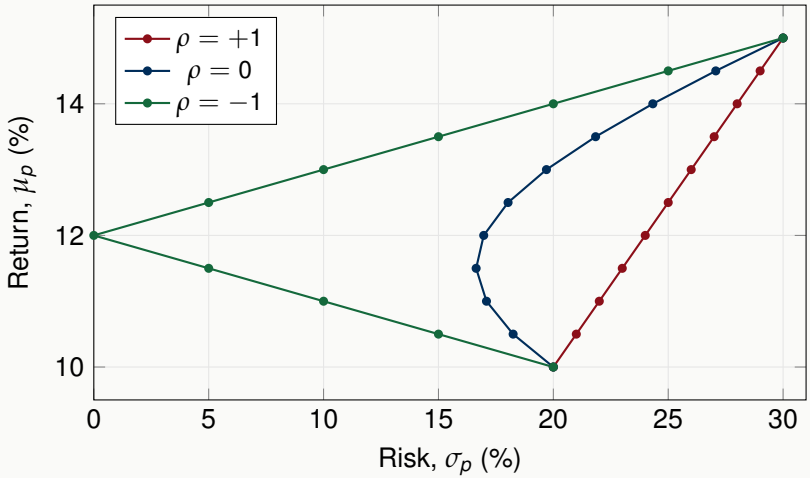
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Lower Correlation Bows the Curve Left

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$\rho = +1$: assets move in lockstep – no diversification benefit at all. As ρ falls toward 0 and then -1, the curve *bows left*: same return, strictly less risk. At $\rho = -1$, some mix even reaches *zero* risk – the umbrella/ice-cream story, drawn precisely.

Many Assets: Matrix Notation

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For $N = 50$ assets: $\binom{50}{2} = 1225$ distinct covariance terms. Writing these by hand doesn't scale.

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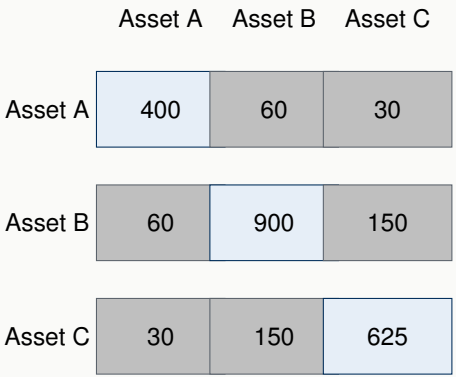
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Check: for $N = 2$, expanding $w^\top \Sigma w$ recovers exactly the two-asset formula.

Visualizing the Covariance Matrix



Diagonal (navy) = variances. Off-diagonal (gray) = covariances, symmetric.

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$w = (0.5, 0.3, 0.2)^\top$ against the Σ above

$$\sigma_p^2 = 0.25(400) + 0.09(900) + 0.04(625) + 2(0.15)(60) + 2(0.1)(30) + 2(0.06)(150)$$

$$= 248 \qquad \Longrightarrow \qquad \sigma_p \approx 15.75\%$$

Systematic vs. Unsystematic Risk

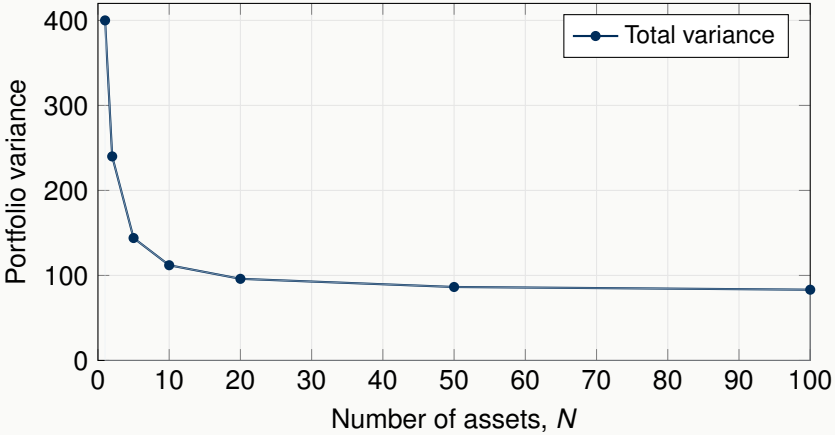
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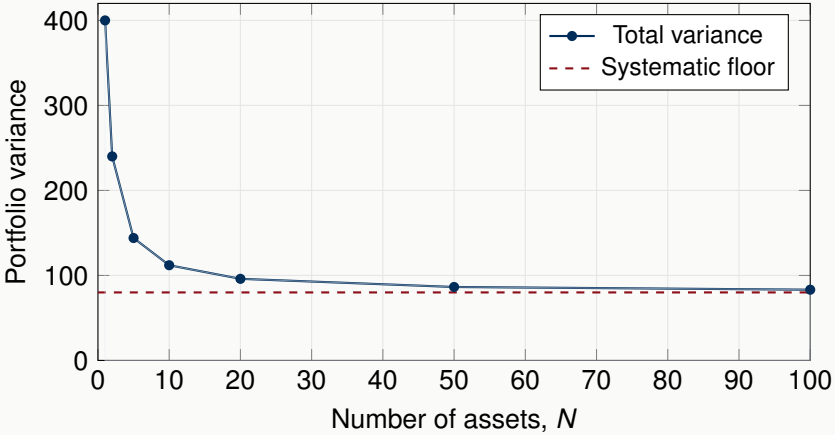
$$\sigma_p^2 = \frac{\sigma^2}{N} + \frac{N-1}{N}\overline{\text{Cov}} \quad \xrightarrow{N\to\infty} \quad \overline{\text{Cov}}$$

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Shaded area = unsystematic risk (diversified away). Dashed floor = systematic risk (remains forever).

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- Lower correlation bows the achievable risk-return set left – less risk, same return.
- As N grows, unsystematic risk diversifies away; only systematic risk remains – the seed of CAPM.

Check Your Understanding

Discuss (2 minutes)

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3. Why must Σ be symmetric?

Looking Ahead

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The Efficient Frontier – Lecture 8